



Faculty of Engineering
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From Cycles to Systems: Designing for Menstrual Health Complexity

Design Team

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Prepared for SYDE 600 | Fall 2024

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Abstract

Menstrual health is a complex systems design challenge that is influenced by biological, psychological, and social factors. However, current solutions often fail to address this problem from a systems-thinking perspective. This design proposal outlines comprehensive user research, including contextual inquiries with diverse stakeholders and participatory design workshops, to incorporate user-driven design for a practical and innovative solution.

Drawing from stakeholder insights, the design team developed a user-centered solution to menstrual health management and ownership that prioritizes physical and educational components, including: a customizable bulletin board-based dashboard with a monthly calendar, including stickers for personalized tracking and tailored scratch-and-learn cards.

Design Overview

Based on the user research presented throughout this paper, the design team has proposed a physical dashboard with the following key features:

1. **Bulletin board base:** Users can seamlessly integrate their existing routines into a physical display with full customizability in terms of placement and tracking ability. Users can control the placement of the board, select which elements are used based on their cycle, and make additions such as whiteboards, reminders notes, and more.
2. **Sticker Sheet Tracking:** Users map flow, energy levels, and symptoms directly on the calendar using colorful, visually striking stickers that are designed to be both appealing and easily distinguishable, making the tracking process engaging and intuitive. These symbols draw attention and enhance the clarity of the information being tracked.
3. **Collectible Scratch-and-Learn Cards:** Attached to the calendar are educational cards, customized to users based on their individual knowledge levels. Users' knowledge is assessed through a brief evaluation when purchasing their calendar board subscriptions. Designed as a daily achievement, these cards help users stay engaged and informed about their menstrual health, fostering continuous learning and empowerment.
4. **Calendar:** The calendar encompasses one general month, providing flexibility and easy customizability. On a subscription basis, users can use the calendar any time of the year, giving them full control and requiring minimal effort to tailor the experience to their personal needs and preferences.



Figure 1. Final Design Concept Developed in Figma

Background

Menstrual health is a dynamic systems design challenge shaped by the interplay of biological, psychological, and societal factors. Unlike static health conditions, it evolves across life stages, influenced by hormonal cycles, cultural norms, and healthcare systems. Current solutions, such as apps and wearables, often focus on surface-level tracking, neglecting systemic issues like stigma, inadequate medical guidance, and variability in user needs. This fragmented approach needs to address menstrual health as an interconnected system requiring adaptable, life-stage-specific solutions.

This project reframes menstrual health as a dynamic ecosystem by adopting a systems design engineering perspective to investigate the problem space and propose a meaningful design solution. The design team collaborated with stakeholders to investigate the interplay between biological processes, behaviours, and socio-technical systems across different life stages, from early adulthood to menopause, identifying the core menstrual health needs through direct engagement. The inquiry process highlighted systemic inefficiencies, including the inadequacy of existing tools, lack of medical support, and barriers caused by misinformation and stigma.

This design endeavour aims to reimagine menstrual health management through a user-centred solution that adapts to evolving needs. By integrating insights from participants and identifying systemic opportunities, it addresses design requirements that meet both immediate challenges and broader socio-technical contexts.

Design Process

The menstrual cycle was a problem space that resonated with the design team as, despite being people who menstruate, they could not claim to be experts on the topic due to the way menstrual health management can vary, be misunderstood, and be mishandled from person to

person. The team's primary motivations for pursuing this investigation were to understand the menstrual health-related challenges experienced by individuals and to identify their deepest needs in order to propose a solution that could effectively and seamlessly support people like themselves. The design process consisted of four key stages: (I) Observation, (II) Data Collection, (III) Design Requirements, and (IV) Participatory Design. A summary of all the stakeholders involved throughout the design process can be found in Table # 1 in the Appendix of this report.

I. Observation Stage

Ahead of approaching stakeholders, the team used a 4 W's (who, what, why and how) approach to outline existing knowledge and guide observations. The results of this initial brainstorming are as follows:

Who? Individuals experience menstruation in all walks of life, experiencing a wide range of symptoms and coping with them in different ways.

What? The topic of menstruation encompasses understanding oneself, knowledge about menstruation, planning around one's cycle, tracking or keeping a history, seeking medical intervention, navigating the medical system, sociological impacts, and much more.

Why? People who menstruate require varying types and levels of support, resources, and tools, and have historically been let down by existing solutions.

How? The design team will uncover key problems by starting with known areas of interest and observing and interviewing people who menstruate across a wide range of experiences

The next step was to identify the key themes and problems that would inform the contextual inquiry plan for data collection. The menstrual cycle is a private, long-term, and often stigmatized phenomenon, making it difficult to observe remotely without causing people to act unnaturally due to insecurity or embarrassment. To maximize the value of observation with minimal interaction, the team avoided leading questions and allowed interviewees to guide the discussion. The team asked users about their symptoms, tracking practices, ageing, coping mechanisms, and available resources.

TABLE I: OBSERVATIONAL INTERVIEW QUESTIONS

#1	Can you describe how your menstrual cycle impacts your mood, energy, and symptoms over a typical month?
#2	Have you used any methods or tools to track your health? If so, what was your experience like? If not, could you share your reasons for not using them?
#3	How, if at all, does your menstrual cycle affect your daily life? How do you understand or respond to any changes you notice in your body?
#4	What resources have you found helpful in responding to changes in your menstrual health (i.e. medical, educational, homoeopathic, therapy)?

#5	What are your expectations about how your menstrual health or hormonal changes may evolve as you age?
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The team conducted 7 observation sessions and uncovered some recurring themes or pain points:

- **Menstrual Health Tracking Tools:** Most participants track their cycles using apps but find the tools lacking in accuracy or convenience. For example, Participant 2 uses the Garmin running watch to track her period alongside her fitness, but does not find the provided fitness recommendations useful to her and ignores them (Fig.2). Participant 4 finds app tracking frustrating due to her irregular cycle and forgetfulness with inputting, showing a desire for a more adaptable tool that would auto-fill gaps of missing information (Fig.3)
- **Symptoms Impact Daily Life:** mood swings, fatigue, cramps, etc. affect user productivity, sociability, lifestyle and daily routines depending on cycle status. Participant 2 claims she is “miserable” in the lead-up to her period due to symptoms, which affects her running performance in the weeks before, a factor that is quite important to her (Fig.2). Participant 4 experiences chronic Bacterial Vaginosis around her period that is difficult to get diagnosed and treated (Fig.3).
- **Gaps in Information:** There is a notable lack of clear, accessible information regarding menstrual health and long-term health. For example, despite reducing her concerning symptoms, having an IUD causes Participant 4 more concern than relief at times when her period disappears and she does not know whether it indicates pregnancy (Fig.3). Some interviewees, such as Participants 6 and 3, felt uncertain and ill-informed about the wide range of symptoms they would face during menopause and were disappointed by the lack of clear guidance from the medical system (Appendix).
- **Inconsistency and Irregularity:** Participant #2 expressed difficulties predicting how her menstrual cycle would behave due to its irregularity, which was a concern echoed in talks with Participant #4 (Fig.2, Fig.3). Additionally, Participant #12 experienced irregular cycle patterns triggered by stressful events, which in some cases prompted her to seek medical attention (Appendix). The inconsistencies in menstrual cycles among interviewees, with some also experiencing very regular patterns, inspired further investigation.

S.E. 21/09/2024, 26 female, PhD student

1. Irregular cycle, difficult to predict due to missing periods (e.g., no period for four months).

- Feels ovulation symptoms like:
 - Extreme bloating ("feel like a whale").
 - More body awareness due to regular running (5-6 months, 4 times a week).
- Focused on how the cycle affects running performance rather than other symptoms.
 - Bad runs or low energy are easier to understand when tied to the menstrual cycle.
- Occasional breast pain, which affects running, but not bothered enough to track it.
- Generally:
 - "Chill" until ovulation.
 - Miserable in the lead-up to menstruation (cravings, bloating, mood swings, low energy).
 - Running becomes more difficult 1-2 weeks before the period.
- Cycle has become more regular recently.

2. Uses a Garmin running watch to track the cycle:

- Inputs the first day of the period, and the watch predicts the rest.
- Default period length is 4 days, but it can be adjusted.
- Tracks heart rate, breathing, steps, and VO2 max, with a focus on cardiovascular health.
- The watch provides recommendations to adjust training based on cycle phases:
 - Resistance to cycle-adaptive training / Rigid training adherenceExample: "Do more cardio to decrease PMS symptoms."
 - Doesn't follow the watch's recommendations
 - Sticks to a rigid, male-oriented training plan.
 - Finds it easier to follow a structured plan and doesn't want to deviate.
 - The watch's suggestions aren't specific enough for her needs.

3. Cravings increase, particularly for sugary or unhealthy food.

- Unsure if cravings are caused by hormones or by consuming sugar itself.
- Worse mood and brain fog.
 - Unsure if brain fog is caused by sugar or hormones.
- Reduced productivity, possibly due to a combination of hormonal changes and poor diet during the cycle.

4. Hasn't found many resources that help with menstrual health changes.

- Tries to reduce sweets in the house but ends up buying them anyway.
 - Sweet cravings are the most prominent symptom.
 - Follows a rigid physical plan regardless of symptoms.
 - Overall health has improved, and the cycle has become more regular due to better diet and exercise.

5. Expects menstrual health to change drastically as she approaches menopause.

- Expects tumultuous hormonal changes, mood swings, and body changes.
- Anticipates negative effects on her training.
- Hasn't thought much about it, as it feels like it's far off (late 40s, early 50s).
- Focuses on being as healthy as possible now:
 - Acknowledges that fitness becomes harder to maintain later in life.
 - Feels better when she exercises regularly.

Highlighting the key pain points:

- **Irregular Cycle & Poor Tracking:** Irregular periods disrupt accurate tracking; average-based predictions are unreliable.
- **Running Impact:** Menstrual symptoms (bloating, breast pain) affect running, but she sticks to a rigid training plan designed for men that doesn't account for hormonal changes.
- **Cravings & Diet:** Strong cravings for sweets before menstruation lead to brain fog and poor productivity, but no clear strategy to manage them.
- **Symptom Tracking:** Tracks only what affects her running (ignores other symptoms like breast pain) due to lack of interest.

Commented [1]: Cycle irregularity / Difficulty in predicting cycle

Commented [2]: Bloating / Body image impact

Commented [3]: Increased physical awareness / Running impact

Commented [4]: Symptom prioritization / Running performance focus

Commented [5]: Low energy correlation / Cycle-linked performance dips

Commented [6]: Selective symptom tracking / Breast pain impact on activity

Commented [7]: Premenstrual symptoms / Mood swings / Energy fluctuations

Commented [8]: Physical impact on exercise / Difficulty in running

Commented [9]: Improved regularity

Commented [10]: Tracking tools / Garmin watch

Commented [11]: Flexibility in tracking tool / Customizable period length

Commented [12]: Health metrics tracking / Cardiovascular focus

Commented [13]: Cycle-based training advice / PMS management suggestions

Commented [14]: Resistance to cycle-adaptive training / Rigid training adherence

Commented [15]: Preference for structured routines / Reluctance to adapt

Commented [16]: Insufficient specificity in tracking tool

Commented [17]: Food cravings

Commented [18]: Mood swings / Cognitive symptoms (brain fog)

Commented [19]: Uncertainty about cravings' source / Hormones vs. diet

Commented [20]: Reduced productivity

Commented [21]: Limited resource availability

Commented [22]: Diet management struggle

Commented [23]: Strict adherence to training / Symptoms disregarded in plan

Commented [24]: Menopause anticipation / Expectation of major changes

Commented [25]: Concerns about training impact / Menopause effects on fitness

Commented [26]: Lack of immediate concern

Commented [27]: Health maintenance focus / Long-term fitness awareness

J.B. 21/09/2024, 21 female, Student

1. Her cycle is irregular, which makes tracking it frustrating. She often forgets to input all the days of her period, leading to incomplete data.
 - Since getting an IUD, her periods have become very mild or sometimes non-existent, but this makes it difficult to distinguish whether missed periods could indicate pregnancy.
 - Prior to the IUD, she used birth control pills but often forgot to take them, despite setting an alarm reminder, which led to using Plan B once. Restarting birth control after that was confusing for her.
 2. She doesn't regularly track her period or symptoms due to the irregularity of her cycle and forgetfulness.
 - Wishes tracking tools would auto-fill her period days to make it easier.
 - Doesn't track symptoms consistently, making it difficult to monitor changes related to her cycle.
 3. Spots frequently, which scared her before getting the IUD. The spotting was one of the reasons she decided to switch to the IUD.
 - Struggles with chronic BV (bacterial vaginosis) that seems to resurface close to her period. It's been difficult to diagnose because it would disappear before she could get a swab from the doctor.
 - Experiences pain after the IUD insertion but doesn't track symptoms systematically.
 4. Hasn't used specific resources for menstrual health or tracking.
 - Considered a supplement and antibiotics for treating BV, but recurrences still happen.
 5. Doesn't know much about menopause or long-term hormonal changes.
 - Hasn't spent much time reading about or preparing for menopause or future changes related to her cycle.
- Commented [41]:** Irregular cycle / Tracking frustration
- Commented [42]:** IUD-related changes / Pregnancy concern due to light periods
- Commented [43]:** Birth control challenges / Forgetfulness
- Commented [44]:** Inconsistent tracking / Cycle irregularity and forgetfulness
- Commented [45]:** Desire for auto-fill in tracking tools
- Commented [46]:** Spotting issues / IUD decision driver
- Commented [47]:** Recurring BV / Linked to menstrual cycle
- Commented [48]:** Post-IUD pain / Inconsistent tracking
- Commented [49]:** Limited resource usage / Menstrual health
- Commented [50]:** Limited menopause awareness / Lack of long-term health planning

Figure 3: Annotated problem observation interview #2 (Participant #4, J.B.)

From the themes and problems identified, the team was able to effectively focus on two high-level areas: (1) existing routines and (2) existing tools. Area 1 emerged from themes of inconsistency, irregularity, and symptoms, while Area 2 stemmed from themes related to information sources and tracking tools. These research areas were also summarized in a flowchart (Fig. 4). Several research goals, inspired by our initial stakeholder discussions, included:

- Understanding the available health tracking tools, as well as their strengths and weaknesses according to users.
- Exploring people's coping methods and how they evolve throughout the menstrual cycle.
- Identifying the health markers most important to individuals and how these are impacted by menstruation.
- Understanding which concerns or irregularities in menstrual health prompt medical intervention and the workflows people follow when seeking help.

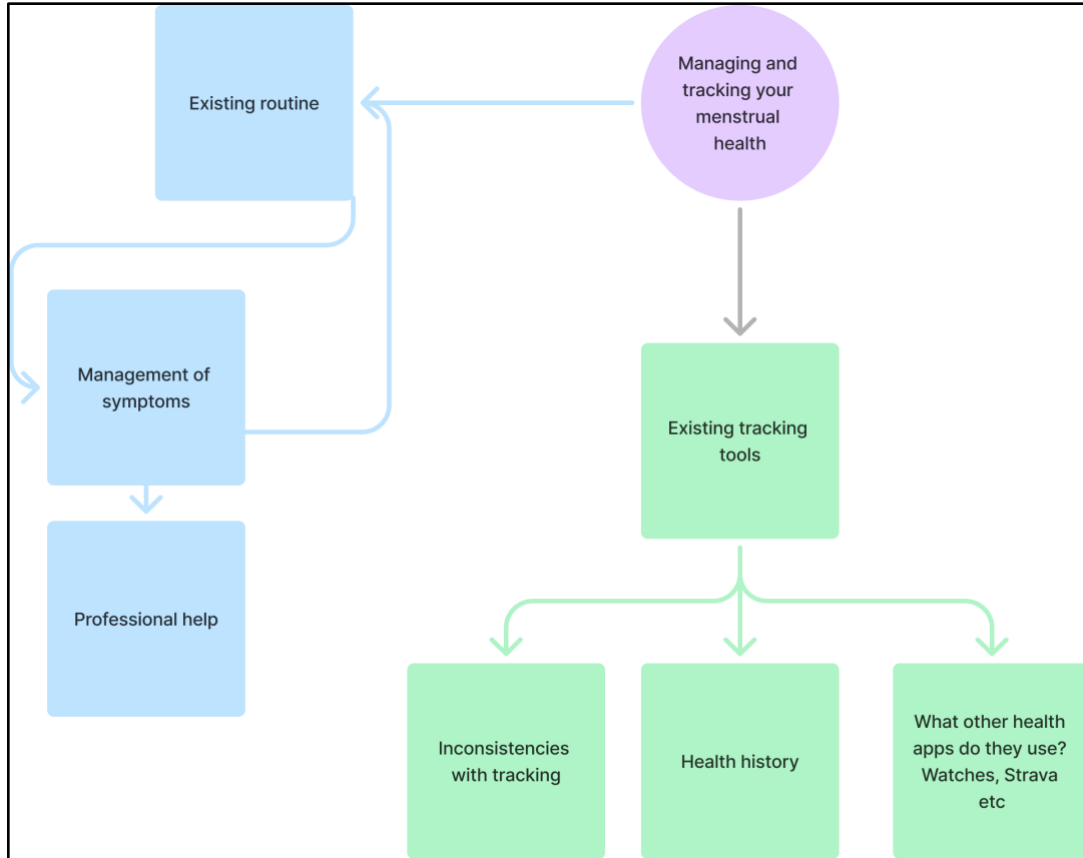


Figure 4. Flowchart of research areas to investigate during data collection stage

II. Data Collection Stage

The methodology employed for data collection was contextual inquiry (CI), a user-centred research method designed to gather insights into behaviors, motivations, and processes related to the menstrual health environment. This method allowed us to develop a deep understanding of individuals' actions, decisions, and challenges surrounding menstrual health by combining observation with direct questioning in a "master-apprentice" teaching style. Giving interviewees more control over the conversation helped reveal issues and needs that would be difficult to uncover in a standard interview setting.

The design team invited eight users and stakeholders who menstruate, aged 18 to 30, to participate in one-on-one interviews. The project's scope was focused on young adults because, during the initial observation stage, significant variations in symptoms, routines, tools, and overall experiences were observed among people who menstruate across different age groups. To create actionable and feasible design requirements for a working solution, it was necessary to concentrate on a specific age group. The interview questions were open-ended to allow participants to demonstrate, teach, and navigate their methods and thought processes (Table 2). Since many users

had easy access to their tools remotely or outside their homes, CI interviews were conducted both over calls and in person at locations where participants felt comfortable. Data was collected through detailed notes, audio recordings for annotation and analysis, and photographs or screenshots of relevant artefacts discussed during the interviews.

TABLE II: FINALIZED CONTEXTUAL INQUIRY PLAN

To observe coping strategies	• Walk me through how you prepare for your period • What symptoms do you care about around your period? ◦ Walk me through how you deal with X symptom • What does a normal day look like at the beginning/middle/end of your period?
To inquire about medical intervention	• What symptoms do you feel are irregular? • Walk me through your process on what to do if X symptom gets out of control ◦ Walk me through the process to seek medical attention
To observe tracking/monitoring tools	• How do you keep track of X symptoms? ◦ Show me how you'd log symptoms on a given day, or ◦ Teach me how you keep track without apps ◦ Looking at X app, why do you use some modules and not others? • How does tracking change throughout the month?
To inquire about health markers	• What health markers are important to you? ◦ Show me how you keep track • How does your period affect this routine?

To identify common pain points and issues experienced by people who menstruate, key themes were derived from the contextual inquiry interviews. First, a consolidated cultural model that included all the engaged interviewees, highlighting the most relevant stakeholders and artifacts, as well as the interviewees' expectations and feelings toward them (Fig. 8). This revealed the commonalities among stakeholders, showing that despite the wide range of experiences among people who menstruate, there were overlapping factors that could help establish common foundations for the design.

The most insightful method of coming to conclusions was a thematic analysis of coded phrases from interview transcripts. These codes were grouped and fed into core themes that defined our problem space (Table 6). For example, Participant #16's comment, "not a fan of doctors" or "of hospitals generally" (Fig.5), and Participant #4's feeling that "no one's giving [her] real

answers” and “no one’s really explained if there are other options” regarding her chronic menstrual health issues (Fig.6) directly contribute to the theme that menstrual health is often misunderstood and mismanaged (Table 6). Regarding menstrual cycle regularity, Participant #4 mentions, “I try to track it, but it doesn’t seem to match up perfectly with my cycle anymore” (Fig.6), and Participant #16 states “I don’t know if my weird symptoms are necessarily normal” (Fig.5), supporting the theme that menstrual cycles are unique and that regularity is subjective. With a solid set of themes outlining the problem space, core design requirements were developed to address in the final stakeholder-centred phase of the design project—participatory design.

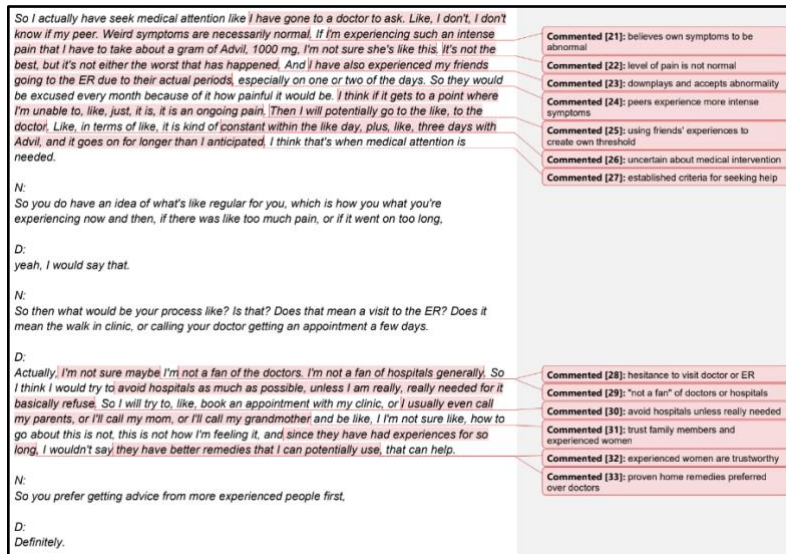


Figure 5. Coded CI interview transcript excerpt #1 (D.K.)

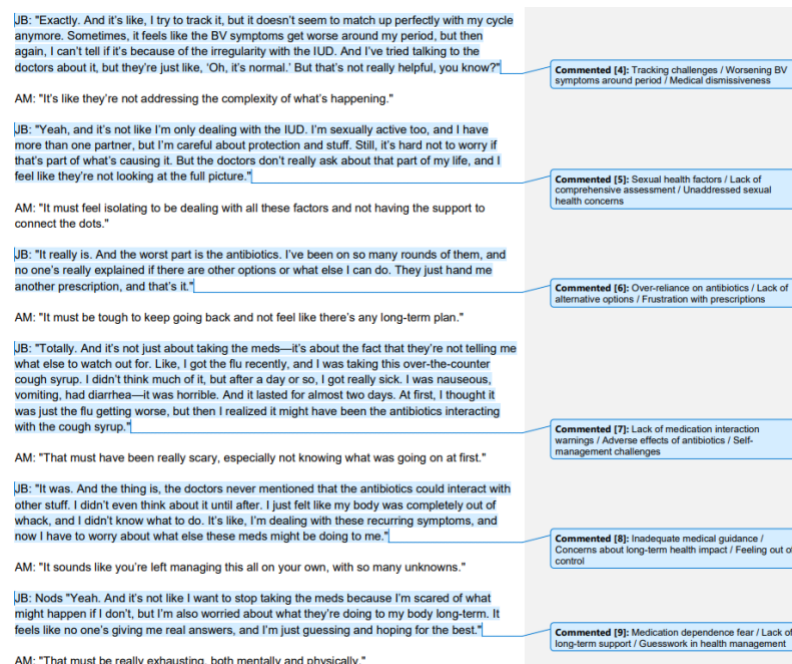


Figure 6. Coded CI interview transcript excerpt #3 (J.B.)

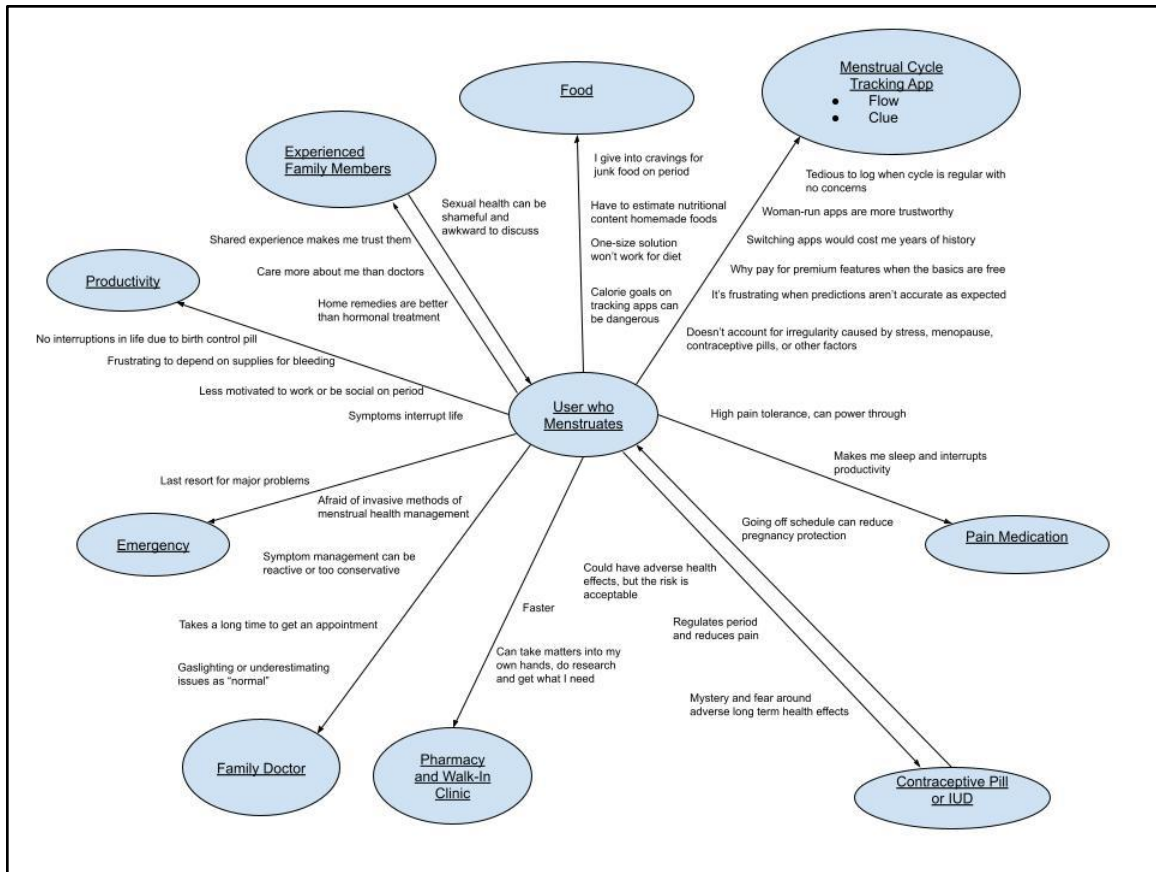


Figure 7. Consolidated CI culture model

TABLE III: CODES AND THEMES FROM CONTEXTUAL INQUIRY

Codes	Grouped Codes/Low-Level Themes	High-Level Themes
See Appendix	Need to create own custom symptom management plan that addresses uniqueness of own cycle	Own health is own responsibility
	Friends and family or self-guided research first resort to addressing concerns in menstrual symptoms	
	Stigma around keeping menstrual health personal or pressure to compare own menstrual cycle to others exists in society	
	Various types of tracking apps exist for collecting history of menstrual symptoms or other health markers important to each individual	
	Need to prepare physical resources for a period	Physical preparedness required
	The rationale and workflow for seeking intervention is to get professional help only in emergencies	Symptoms are normalized and dismissed
	Major physical pain that interrupts life is normalized	
	Outliers: For some, pain is minimized or minor and does not affect daily life	
	Prescribed contraceptives are misunderstood but the trade off between positive and negative health effects is accepted	Menstrual care is misunderstood and mismanaged
	The medical system can fail due to a lack of care or understanding of unique menstrual cycles	
	Hesitation around medical intervention and lack of quality of care from the health system	
	Mental well-being fluctuates over the menstrual cycle which impacts motivation and energy	Menstrual cycles are unique and regularity is subjective
	Tracking can help keep a history and make predictions for expected symptoms - users desire accurate information	
	Active tracking is undesirable when symptoms are regular or predictive information is not useful	
	Convenience and regularity in routine is favoured, and menstrual cycle can cause interruptions in routine for some	Convenience and unbroken routines are favoured
	Drawbacks with apps are accepted due to the inconvenience of making changes	

III. Participatory Design

To initiate the design process, the core themes identified during the data collection stage were transformed into actionable design requirements. This involved reorganizing and refining the high-level themes into specific criteria that a potential design solution would need to address. Using the six design requirements outlined in **Table #4**, a protocol was developed to engage stakeholders in the participatory design (PD) process. This approach ensured that the solution met the true needs of stakeholders, enhanced the likelihood of acceptance, and facilitated seamless integration into the lives of the majority.

TABLE IV: DESIGN REQUIREMENTS

Themes	Design Requirements (the design should ...)
1. Own health is own responsibility	Allow the user to independently monitor and understand their health
2. Physical preparedness required	Promote physical preparedness related to one's menstrual cycle and support in planning for upcoming physical changes
3. Symptoms are normalised and dismissed	Accept and understand one's symptoms instead of normalizing and dismissing
4. Menstrual care is misunderstood and mismanaged	Provide user education on menstrual health to help users understand and manage changes
5. Menstrual cycles are unique and regularity is subjective	Embrace uniqueness by accommodating individual variability in menstrual cycle patterns
6. Convenience and unbroken routines are favoured	Seamlessly integrate into users' existing routines without disruption

The protocol for the PD sessions was inspired by a paper by Bardzell et al. (2019), which covered the topic of menopause and included sessions that explored various scenarios about women's health while generating potential ideas to address the associated challenges. This approach proved particularly instrumental in the final stages of developing a solution, where it was

implemented as a method of translating stakeholder input into tangible design concepts. Prior to stakeholder engagement, the design team conducted an internal pilot test of the PD plan to identify potential issues and develop sample designs to inspire participants. Additionally, in anticipation of varying levels of enthusiasm or creativity from stakeholders, the team developed strategies to encourage participation and foster engagement. Through this pilot test, the team concluded that drawing with coloured pens and verbal discussions were effective tools for offering sufficient freedom for stakeholders to visualize potential solutions.

The finalized PD protocol involved three stages: (1) Ranking and grouping of design requirements, (2) Brainstorming ideas, and (3) Sketching ideas. Ranking and grouping the design requirements into "nice-to-have" and "must-have" categories ensured that users were familiarized with the requirements and focused on their most prioritized features, providing clear direction for their designs. During this first stage, stakeholders were encouraged to ask any questions related to the requirements in order to clarify any confusion. The second stage involved brainstorming ideas to address the design requirements that stakeholders had ranked as most important. To avoid relying on conventional digital solutions, which could potentially restrict user autonomy by placing dependence on software developers, the team emphasized the importance of brainstorming physical or social solutions. Session participants were provided with resources (e.g., paper, coloured pens, sample solutions) to sketch their ideas, helping them better visualize their concepts while articulating how their solutions would function. A clear outline of the protocol is in **Table # 5**.

TABLE V: PARTICIPATORY DESIGN (PD) PROTOCOL

- | |
|---|
| <ul style="list-style-type: none"> A. Rank the proposed requirements from least important to most important. B. Group the requirements in 2 groups: nice to have or need to have? C. Discuss ideas with participants (~15 min). <ul style="list-style-type: none"> a. The solution can be physical, social, or digital (emphasis on physical or social). b. Propose ideas for each requirement prioritized by the participant. D. Select a few ideas to sketch out and/or create in detail (~15 min). <ul style="list-style-type: none"> a. Constraints: Avoid a digital solution. |
|---|

Five stakeholders were individually engaged in in-person design sessions using the PD protocol. Participants recorded their thoughts on the design requirements and potential solutions through drawings and recorded discussions. For stage 1 of the protocol, design requirement rankings were generally unique between stakeholders, however, design requirement #1 and #5 had the best average ranking values with 2.4 and 2.8, respectively. Alternatively, design requirement #4 had the worst average ranking value of 4.6. Interestingly, participant C.H. ranked the design requirements against the observed trend for design requirements #1 and #4 (i.e., ranked #1 as 6 and #4 as 1).

In the decision between "need-to-have" and "nice-to-have," users categorized requirements with significant diversity, with few clear trends identified. However, it was noted that design

requirement #5 was categorized as "need-to-have" by all stakeholders, which corresponded with it having the second highest priority in terms of rankings.

The results of the PD highlight the uniqueness in stakeholders' rankings and categorizations, emphasizing the need to address popular requirements like #1 and #5. However, the lack of convergence in this stage of the PD session emphasized the need to incorporate flexibility into the design process, allowing for customization and fostering a sense of ownership among users to ensure the solution meets diverse needs. A table displaying the average rankings and categorizations of the design requirements are displayed in **Table # 6** below:

TABLE VI: RANKING AND CATEGORIZING OF THEMES

#	Themes	Rankings <i>(1 = high importance, 6 = low importance)</i>	Categories <i>(1 = "need to have", 0 = "nice to have")</i>
	Allow the user to independently monitor and understand their health	1, 1, 3, 1, 6 (Average = 2.4)	1, 1, 0, 1, 0 (Average = 0.6)
	Promote physical preparedness related to one's menstrual cycle and support in planning for upcoming physical changes	3, 4, 2, 6, 5 (Average = 4.0)	1, 0, 1, 0, 0 (Average = 0.4)
	Accept and understand one's symptoms instead of normalizing and dismissing	6, 2, 4, 4, 4 (Average = 4.0)	0, 1, 0, 1, 1 (Average = 0.6)
	Provide user education on menstrual health to help users understand and manage changes	5, 6, 6, 5, 1 (Average = 4.6)	0, 0, 0, 1, 1 (Average = 0.4)
	Embrace uniqueness by accommodating individual variability in menstrual cycle patterns	4, 3, 1, 3, 3 (Average = 2.8)	1, 1, 1, 1, 1 (Average = 1.0)
	Seamlessly integrate into users' existing routines without disruption	2, 5, 5, 2, 2 (Average = 3.2)	1, 0, 0, 1, 1 (Average = 0.6)

**Note: Rankings or categories in the same list indices are recorded from the same stakeholder. For example, the last ranking indices (6,5,4,1,3,2) are from participant C.H. The order of the stakeholders in the rankings columns are S.K., M.S., M.H., E.S., C.H.*

In stage 2 of the PD sessions, the majority of ideas brainstormed by participants were physical solutions, such as physical calendars or wearable point-of-care temperature/pressure-dependent sensors. Additionally, one stakeholder suggested creating a platform where people who menstruate could share their personal experiences with symptoms or issues related to menstruation, offering information on how to address these problems. For digital solutions, other stakeholders expressed preferences for apps based on frustrations with existing apps, such as notifications at specific points in their cycle or a hands-free app that emphasizes minimal user interaction. **Table #7** summarizes the ideas generated by stakeholders.

TABLE VII: SUMMARY OF IDEAS GENERATED BY STAKEHOLDERS

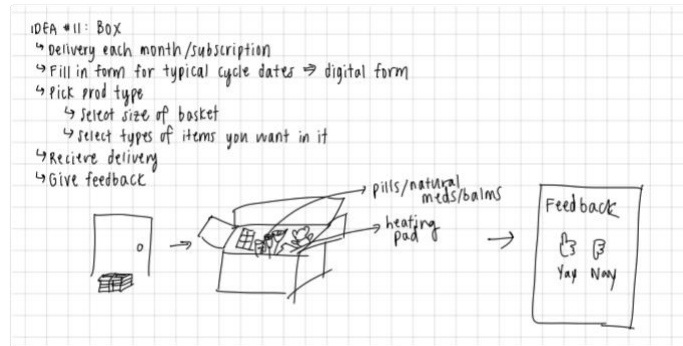
Stakeholder	Ideas	Solution
S.K. M.H.	Items: Physical Calendar, Device/Clicker, Posters in CC, Alarm clock (measures cycle), Stuffed animal, Period basket Beautifully designed calendar or artistic work for tracking mood, symptoms, and patterns (utilizes markers/stickers, is engaging).	Physical
S.K. C.H. M.S.	Heating pad with UI to track. Heating/warm bottle Cooling/heating pillow that can adjust temperature based on body temperature. Helps to maintain a stable body temperature. Stylish shawl with advanced fabric to maintain stable body temperature (addresses hot flashes or chills).	
S.K.	Services: Testing center (takes blood, pp simulator), Spa (for those on period), Delivery service (treats, flowers, natural products), Call center, 23 and Me (for period)	
M.S.	Stylish bracelet that measures biomarkers tied to menopause. Can give indicators when hot flashes occur or when hydration is needed.	
M.S.	Heat/pressure sensitive stones that change colour discreetly. Indicators to help with self-awareness and promote relaxation.	

M.S.	Record physical symptoms by dropping coloured stones corresponding to certain symptoms into a jar. Visually tracks symptoms felt over time.	
M.H.	Cards with tips, affirmations, and exercises tailored to the user.	
E.S.	Custom exercise plan that is aligned to the users' cycle and exercise goals.	
S.K.	Hire nanny	Social
E.S.	Social platform for sharing and seeking information on symptoms or issues. First-hand accounts and experiences aid users in finding community and advice.	
S.K. C.H. E.S.	App - better clue App (UFlow) - Send notifications about different stages of cycle. Create an app that changes with the user life stages. Adaptive app with emphasis on minimal need for interacting with applications. App has a comprehensive overview of users' cycles and sends notifications as needed to minimize disturbing the user.	Digital
S.K.	Product rec systems	

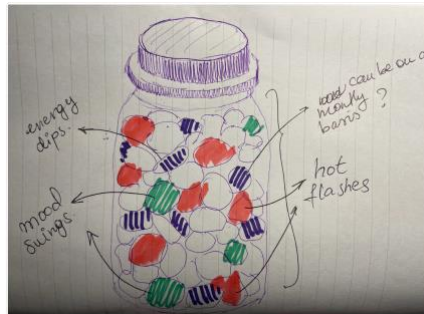
Stage 3 of the PD sessions resulted in several sketches, mood boards, and descriptions of solutions ideated by stakeholders to address their most important design requirements. **Table #8** below presents the key solutions on which the design team's final proposal is based.

TABLE VIII: DRAWN OR DESCRIBED SOLUTIONS

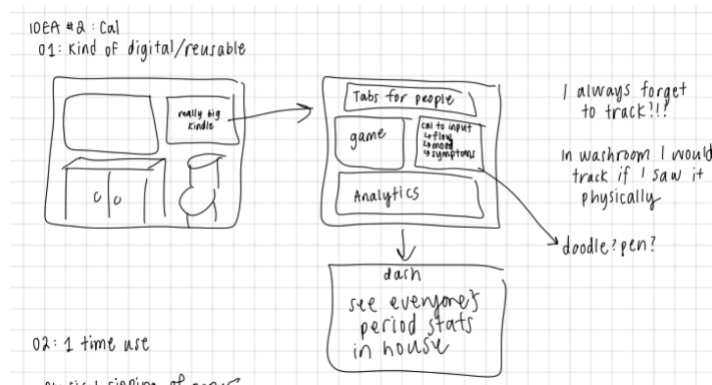
Delivery Box: A subscription-based delivery box product that allows users to customize their gift basket with items. Users are allowed to specify what size of basket and items they want in it. Additionally, users can input cycle data to ensure that the basket arrives on the correct date. Users can provide feedback to enhance product customization to their specific needs.



Symptoms Jar: Users can record their symptoms felt during a cycle or time by mapping symptoms to different coloured stones. When symptoms are felt, users can drop a stone into the jar. At the end of the cycle/period, users can quickly evaluate what their most prevalent symptoms were by assessing which colours are displayed in the majority.



Digital Dashboard Stat Tracker: A digital dashboard that can be placed anywhere to display tracked statistics and analytics on the user(s) health status. Additionally, it is interactable and can be utilised to track symptoms and other data.



Proposed Design and Recommendations

After conducting the participatory design sessions, the design team used the insights gained to inform and develop the final design. Taking into account the stakeholder recommendations from previous sessions, the team presented all ideas in a 2x2 matrix format, and evaluated the number of requirements met and the physicality of the solution. The matrix headers were labelled “no requirements” to “all requirements” on the y-axis and “digital” to “physical” solutions on the x-axis (see Figure 8). Considering that various digital solutions, such as Apple’s health tracking app - ad Clu, have already addressed the digital solution space, the team decided to focus on a physical design solution to disrupt the market and innovate in an area that has not been the main focal point in previous research. This allowed the team to focus on the top-right quadrant for the proposed design as these physical solutions fulfilled the majority of design requirements outlined in the participatory design sessions (see Table 9).

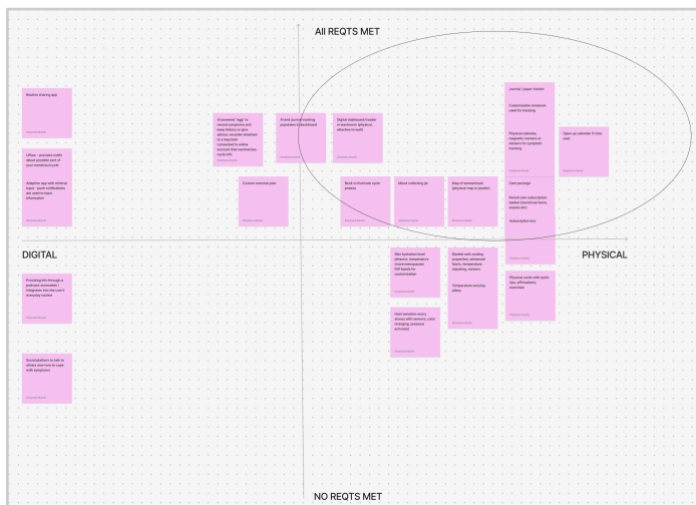


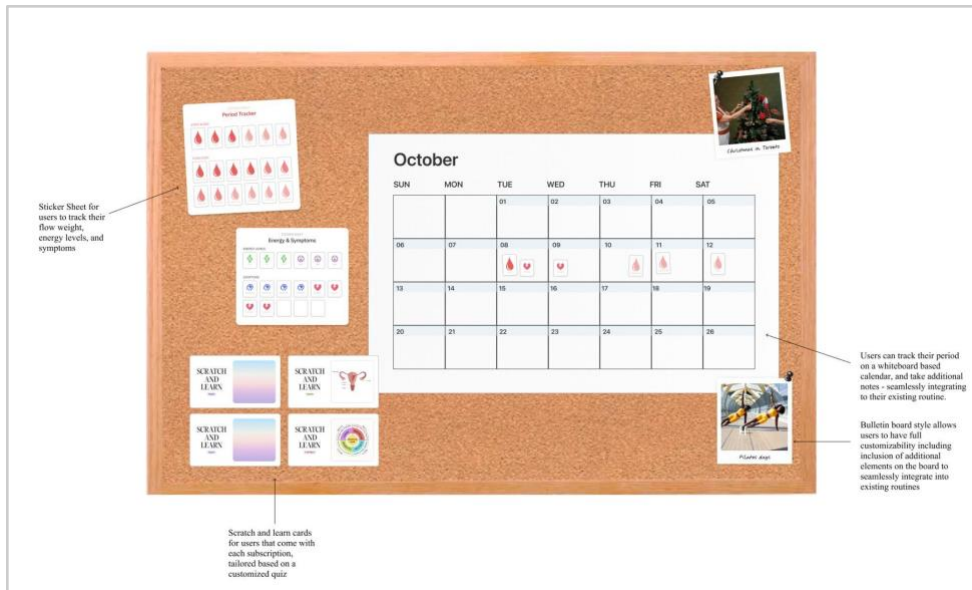
Figure 8. 2x2 Design Matrix

TABLE IX: STAKEHOLDERS AND IDEAS FROM PARTICIPATORY DESIGN

The following are ideas presented in the top-right quadrant:

Stakeholder	Ideas
Tracking	
S.K.	Digital dashboard/tracker in the washroom (physical, attaches to the wall)
M.S, S.K.	Physical calendar, magnetic markers or stickers for symptom tracking
S.N.	Journal/paper tracker
S.K.	Digital dashboard/tracker in the washroom (physical, attaches to the wall)
Educational Resources	
S. N.	Map of womanhood (physical map or poster)
C.H.	Book to illustrate cycle phases
Mood and Wellness Monitoring	
N.S.	Customizable notebook used for tracking
S.N., S.T., S.K.	Period care subscription basket (menstrual items, snacks etc)
S.N.	Care package, subscription box
M.S.	Mood collecting jar

After identifying the most valuable ideas, the team designed the following concept displayed in Figure 1, beginning with a sketch and wireframe. Figma was used to digitize the idea into a higher-fidelity prototype (Fig.1). This includes the following features:



1. Bulletin board-based dashboard

- Medium or large sized board with cork, whiteboard, and card slots
- Hung with removable, strong sticker hooks
- Any staples, pushpins, and whiteboard markers can be used for flexibility

2. Sticker Sheet Tracking

- Icon sticker sheets separately provided for markers including: mood, energy, period, pain, sleep, contraceptives, menopausal symptoms, workouts, cravings, etc.
- Compatible with continuous improvement: additional icons to be created based on future stakeholder engagement through PD.

3. Collectible Scratch-and-Learn Cards

- Card slots filled with custom colour-coded educational cards to be engaged with frequently by the user
- Users' knowledge is assessed through a brief evaluation when purchasing their calendar board subscriptions
- Topics include: menopausal preparation, strategies for acceptance and normalization of menstrual cycles, improving sleep, improving exercise based on cycle, and more.
- Information to be sourced, and open to continuous improvement.

4. Calendar

- Calendar designed as one general month for flexibility
- Calendar design theme customizable by the user.
- Customizable by stickers, and space for users to write in their own notes.
- Use of stickers allows for future digitization efforts.

Discussion

I. Overview of Results and Key Findings

This project explored menstrual health as a dynamic, multi-faceted challenge, shaped by individual variability, stigma, and gaps in existing tools and education. Through extensive user research and participatory design workshops, the most critical needs identified were the ability for users to monitor their health independently and tools that adapt to the unique and subjective nature of their menstrual cycles. These priorities—**own health is own responsibility (requirement #1)** and **menstrual cycles are unique and regularity is subjective (requirement #5)**—were consistently ranked as the most important by participants and guided the development of the final design.

The proposed solution is a subscription-based physical product including a customizable dashboard, sticker sheet tracking system, and tailored educational scratch-and-learn cards. Each component directly addresses the identified priorities, ensuring a user-centered approach to menstrual health management.

How the Design Addresses "Own Health is Own Responsibility" (Requirement #1)

The ability to independently monitor and understand menstrual health was the most consistently prioritized requirement. Participants expressed frustration with existing solutions that either required reliance on medical systems or offered limited personalization. The **customizable bulletin board-based dashboard** empowers users to take control of their health management by providing a flexible, physical tool they can adapt to their daily routines. Whether mounted on a wall or used on a desk, this dashboard integrates seamlessly into the user's environment, allowing them to track symptoms, energy levels, and other key health markers in a way that feels natural and intuitive. The **sticker sheet tracking system** further supports independent monitoring by simplifying the process of recording and visualizing health trends. Users can track their data without needing to engage with complex apps or systems, fostering a sense of autonomy and reducing reliance on external tools.

How the Design Addresses "Menstrual Cycles are Unique and Regularity is Subjective" (Requirement #5)

The uniqueness of menstrual cycles and the variability of symptoms were highlighted as critical factors to address. Participants often found existing tools too rigid, failing to accommodate irregular patterns or evolving health needs. The **sticker sheet tracking system** is designed for flexibility, allowing users to customize how they document their symptoms, energy levels, and flow. By providing users with the ability to define their tracking parameters, the system respects individual differences and avoids imposing a one-size-fits-all solution. Additionally, the **scratch-and-learn educational cards** are tailored to the user's current knowledge level, assessed through a quick evaluation when they subscribe. These cards adapt to the user's changing needs over time, offering relevant information that evolves with their menstrual health journey.

II. Addressing Other Themes

"Symptoms are Normalised and Dismissed": The educational cards directly address the normalization and dismissal of symptoms by validating users' experiences. These cards offer tailored, actionable insights that encourage users to engage with their health proactively rather than ignoring or dismissing important symptoms.

"Menstrual Care is Misunderstood and Mismanaged": The gap in reliable menstrual health education contributes to poor management of symptoms and care. The scratch-and-learn cards help bridge this gap by providing tailored, accessible guidance. However, broader issues, such as inconsistent medical support, remain outside the scope of this design and represent areas for future exploration.

"Physical Preparedness Required": The design indirectly supports preparedness by enabling users to track and anticipate symptoms. However, explicit tools for preparedness—such as curated product kits or resources addressing specific symptoms—are missing. Future iterations could incorporate features like these to better meet this requirement.

"Convenience and Unbroken Routines are Favoured": Both the dashboard and the subscription model are designed with convenience in mind, ensuring they integrate smoothly into users' existing routines. The physical nature of the tools reduces reliance on digital technology, appealing to users who prefer hands-on, low-maintenance solutions.

III. Implications for Key Pain Points

By addressing these priorities and themes, the design empowers users to better understand and manage their menstrual health. It bridges the gap between the need for personalized, flexible tools and the convenience of physical, user-friendly solutions. The approach not only simplifies health management but also fosters a sense of ownership and confidence in users' ability to handle their unique challenges.

IV. Limitations of the Study and Prototype

Despite its strengths, the solution has some limitations. The focus on physical tools may limit appeal to tech-savvy users accustomed to digital solutions, and the demographic scope of the research (primarily young adults) excludes insights from older populations, especially those approaching menopause. Additionally, while "physical preparedness" and "misunderstood and mismanaged care" were partially addressed, there is room for further innovation in these areas.

V. Next Steps

The design team's plan moving forward includes several key steps to further refine and expand the solution:

- Develop a Digital Counterpart: Creating a web-based platform to complement the physical tools, providing users with a seamless way to manage subscriptions and access advanced tracking features online.

- Conduct User Testing: Re-engaging with the original participants to validate the prototype and gather real-world feedback, ensuring the solution aligns with user needs and expectations.
- Expand the Demographic Scope: To make the design more inclusive, the team aims to involve older users and other diverse stakeholder groups, broadening the applicability of the solution.
- Iterate on the Design: Based on user feedback and additional insights, the team will address gaps in "physical preparedness" and "misunderstood and mismanaged care." This may include incorporating curated kits or collaborating with healthcare providers.
- Prepare for MVP Launch: Finally, the design team will focus on finalizing a scalable and inclusive minimum viable product (MVP) for initial market testing, laying the groundwork for broader implementation.

By prioritizing independent health monitoring and adaptability to unique cycles, the proposed solution offers a tangible, user-centered approach to menstrual health management, meeting the most pressing needs identified by users. This foundation supports future innovation and scalability.